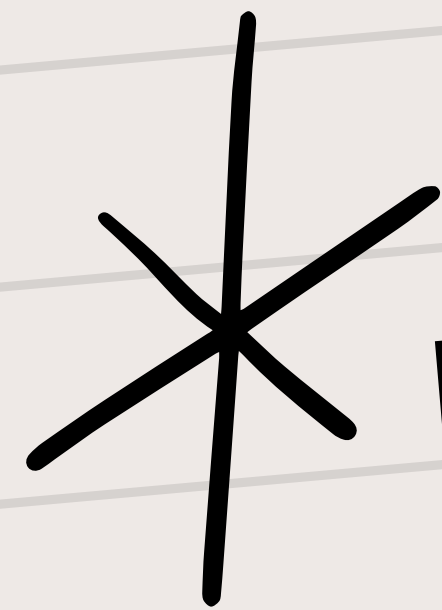
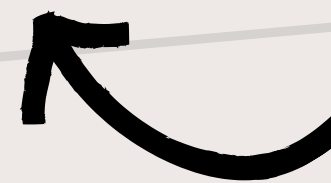
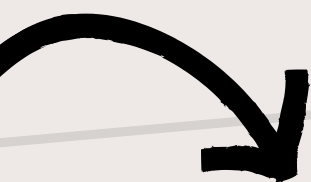




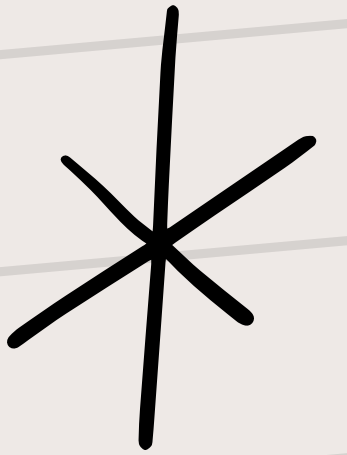
RT-IoT2022 Dataset Analysis

by Amora & Dano



What is RT-IoT2022?

- A real-time IoT network traffic dataset
- Contains 123,117 instances and 85 features
- Includes normal traffic and 12 attack types (DoS, DDoS, PortScan, etc.)



Goal of Analysis:

- Traffic patterns (protocols, services, duration)
- Packet rates and sizes
- Inter-arrival times
- Network flag counts



*What We Did

EXTRACT

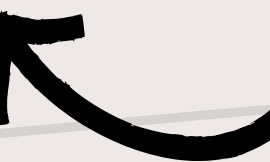
- Loaded the dataset (RT_IoT2022.csv) into Python using pandas

LOAD

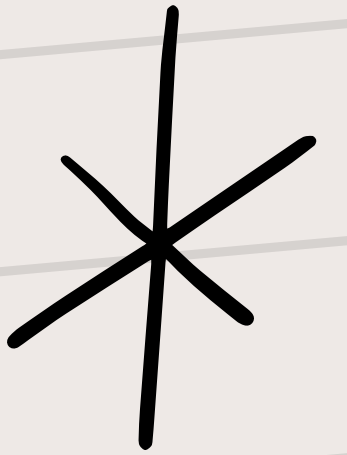
- Saved cleaned data for reuse
- Performed statistical analysis and visualizations to answer all guide questions

TRANSFORM

- Checked for missing values - none found
- Removed duplicates - none found
- Verified data types and cleaned column names



Insights 1

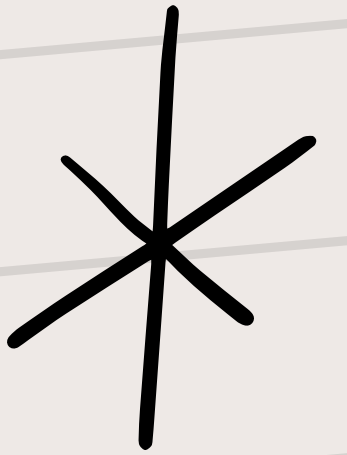


What we Found:

- Most of the data is DOS_SYN_Hping (77%) - everything else is much smaller
- TCP is everywhere, especially in DoS and port scans
- UDP shows up mostly in NMAP_UDP_SCAN and Thing_Speak
- Normal MQTT traffic? Only in the MQTT_Publish class
- SSH only appears when someone's trying to brute force it
- DNS, HTTP, SSL? Mostly in Thing_Speak



Insights 2

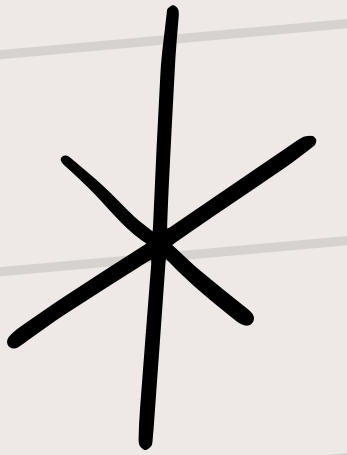


What we Found:

- Flow duration: Wipro_bulb longest (586 sec) DoS and NMAP scans very short near zero
- Statistical test: ANOVA p-value < 0.001 – differences are significant
- Inter-arrival times: NMAP scans have very low IAT (rapid packets) normal MQTT has higher IAT
- The Attack traffic is faster and shorter than normal traffic



Insights 2



What we Found:

- Payload size: Negative correlation (-0.30) – attacks use smaller packets
- Packet rates: Negative correlation (-0.25) – attacks send faster
- Flag counts: SYN flags (0.21) and ACK flags (0.20) most indicative of attacks
- Small packets and high SYN flags will have strong attack indicators



Recommendations

- **Monitor SYN and ACK flags** – spikes usually indicate attacks
- **Watch for small payloads** – attacks consistently use tiny packets
- **Set IAT thresholds** – very fast packets suggest scanning activity
- **Create protocol-specific rules** – watch UDP traffic to DNS/HTTP
- **Use key features** – flags, payload size, packet rates for detection models

